5

159.302 CSPs

Introduction

Constraint Satisfaction Problems

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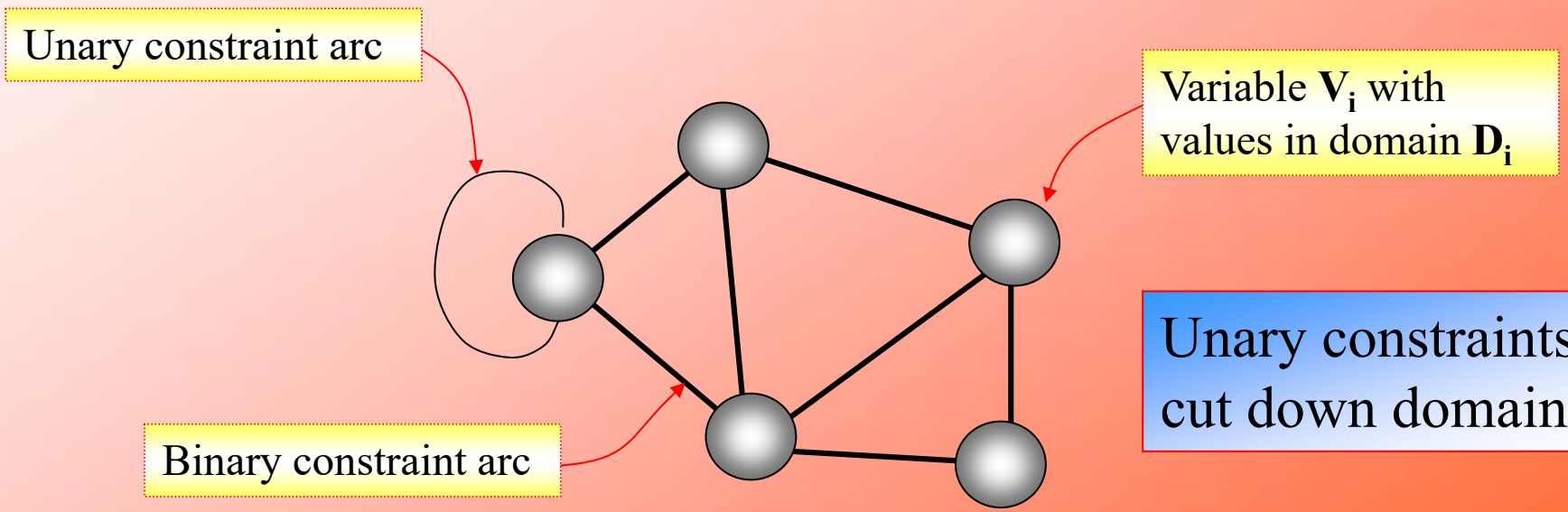
Reference of contents: ***MIT OpenCourseWare***

CSP

General class of problems: BINARY CSP

Application areas of CSPs:

- scheduling tasks, robot planning tasks, puzzles, molecular structures, sensor interpretation tasks, etc.



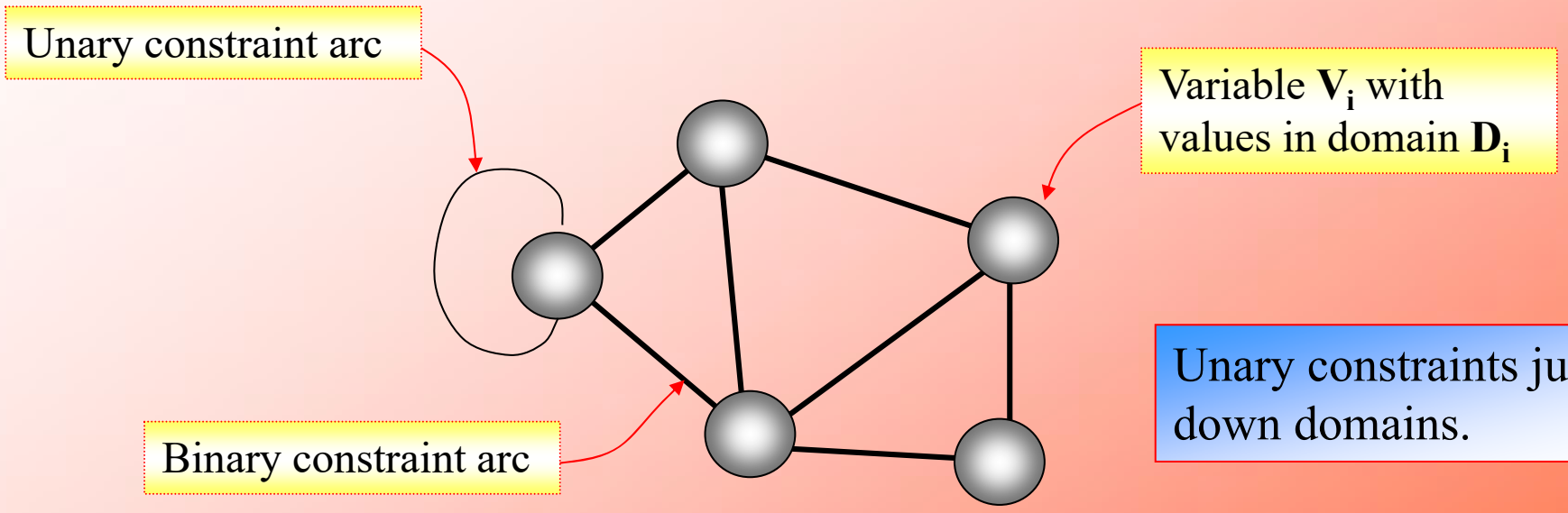
Unary constraints just cut down domains.

This diagram is called a **constraint graph**.

1. **Variables**
2. **Domain of variables**
3. **constraints**

CSP

General class of problems: BINARY CSP



This diagram is called a **constraint graph**.

Basic problem:

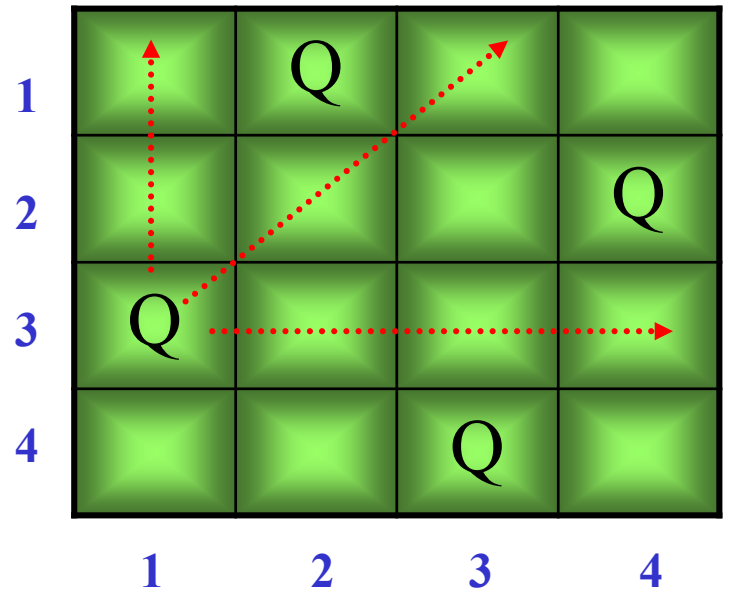
- Find a $d_j \in D_i$ for each V_i s.t. all constraints are satisfied

(finding consistent labeling for variables)

N-Queens as CSP

Classic "benchmark" problem

Place N queens on an N × N chessboard so that none can attack the other.



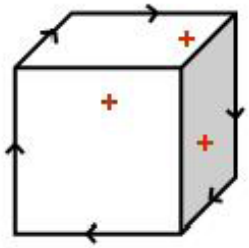
Variables are board positions in N × N chessboard

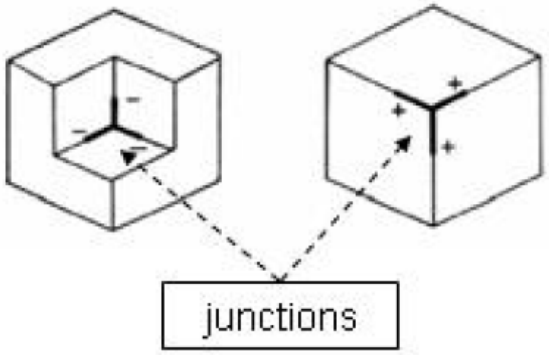
Domains Queen or blank

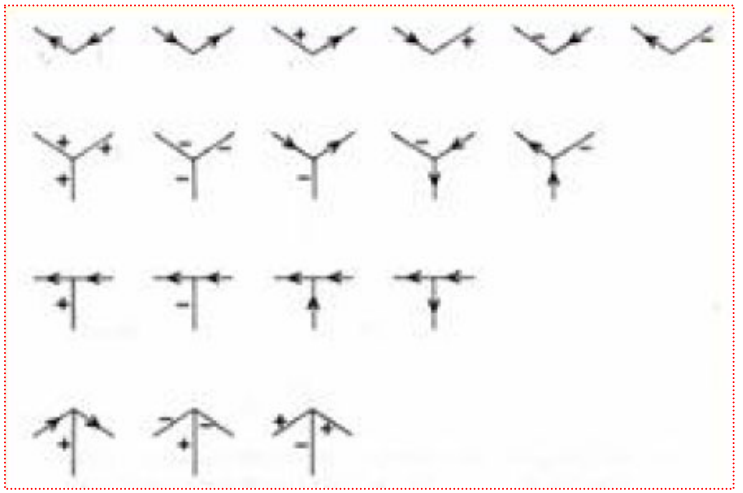
Constraints Two positions on a line (vertical, horizontal, diagonal) cannot both be Queen

Line labelings as CSP

Labeling lines in drawing as convex (+), concave (-), or boundary (>).







All legal junction labels for four junction types.

Variables are line junctions

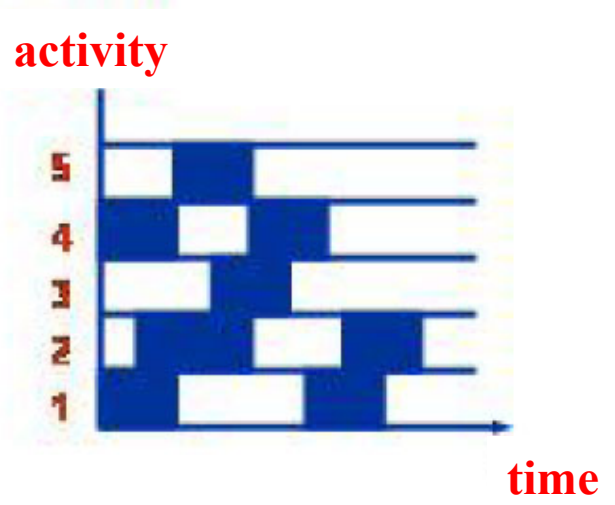
Domains are set of legal labels for that junction type

Constraints shared lines between adjacent junctions must have the same label.

CSP

Scheduling activities that share resources as CSP

Choose time for activities (e.g. observations on Hubble telescope, or semesters to take required classes).



Variables are activities

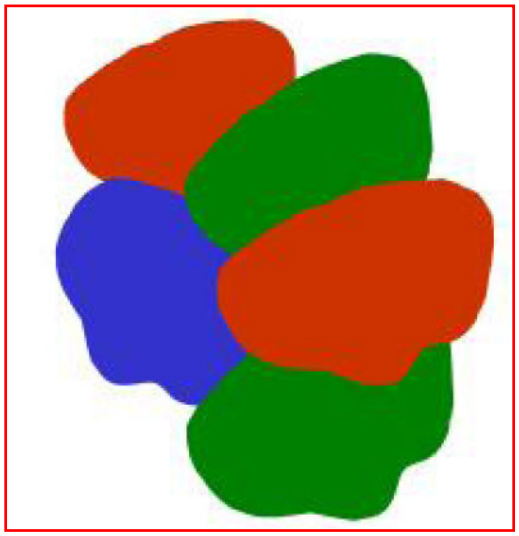
Domains are sets of start times (or "chunks" of time)

Constraints

1. Activities that use same resource cannot overlap in time.
2. Preconditions satisfied.

Graph Colouring as CSP

Pick colours for map regions, avoiding colouring adjacent regions with the same colour.



Variables are regions

Domains are colours allowed

Constraints adjacent regions must have different colours

3-SAT as CSP

Boolean Satisfiability problems - the original NP-complete problem

Find values for boolean variables A, B, C, … that satisfy the formula (in conjunctive normal form).

(A or B or !C) and (!A or C or B)

1-in-3-SAT

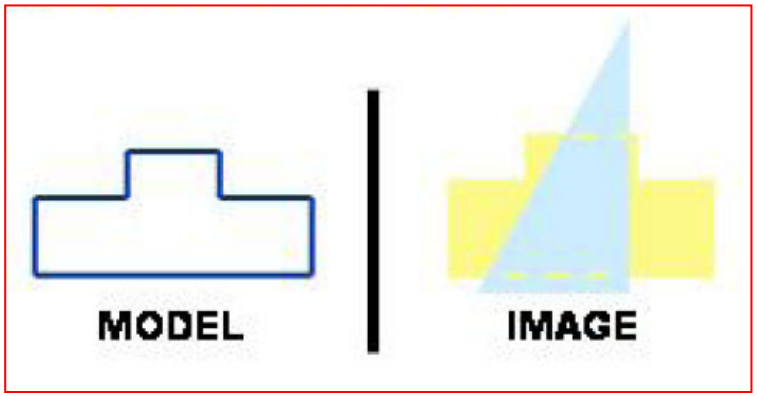
A variant of 3-SAT; determine whether there exists a truth assignment to the variables so that each clause has exactly one TRUE literal

Variables	are clauses
Domains	boolean variable assignments that make the clause true
Constraints	clauses with shared boolean variables must agree on value of variable.

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Model-based recognition as CSP

Find given model in edge image, with rotation and translation allowed

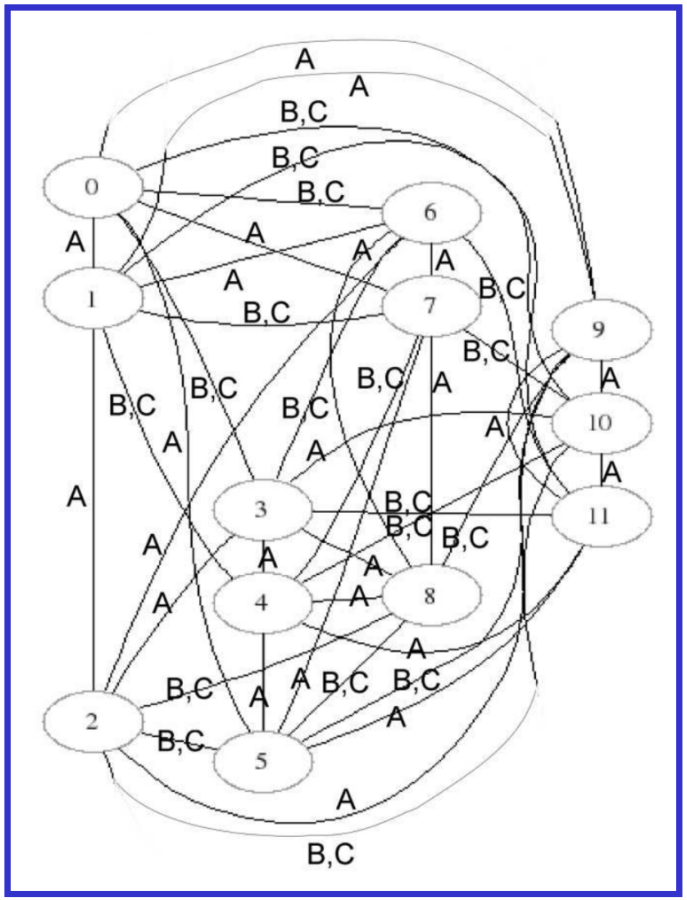


Variables are edges in model

Domains set of edges in image

Constraints angle between model & image edges must match

3D Geometric Object Recognition: Box



Constraint Graph

Variables: 12 line segments in the model

Domain: current image segments

Constraints:

- A = line segments are connected
- B = line segments are parallel
- C = line segments are of similar length



Fig. 1. After segment detection



Fig. 2. Object recognition results

Kasprzak et al, A constraint satisfaction framework with Bayesian inference for model-based object recognition

https://www.gecode.org/



generic

constraint

development

environment

Gecode

Gecode is an open source C++ toolkit for developing constraint-based systems and applications. Gecode provides a constraint solver with state-of-the-art performance while being modular and extensible. Gecode is:

Good News / Bad News

Good News very general & interesting class problems

Bad News includes NP-Hard (intractable) problems

So, good behaviour is a function of domain and not the formulation as CSP.

CSP

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Example

Given **32 papers** (159101, 159201, …, 159334, 159302) & **8 semesters** (Fall 1, Winter 1, …, Winter 4). Find a legal schedule.

Assume that we are given a list of all papers available for the curriculum, together with their pre-requisites requirements, semester offered and time offered.

CSP

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Example

Given **32 papers** (159101, 159201, …, 159334, 159302) &
8 semesters (Fall 1, Winter 1, …, Winter 4).
Find a legal schedule.

Assume that we are given a list of all papers available for the curriculum, together with their pre-requisites requirements, semester offered and time offered.

Constraints

- Pre-requisities
- Papers offered on limited semesters
- Limited number of papers per semester
- Avoid time conflicts

CSP

Example

Given **32 papers** (159101, 159201, …, 159302) & **8 semesters** (Fall 1, Winter 1, …, Winter 4). Find a legal schedule.

Constraints

- Pre-requisities
- Papers offered on limited semesters
- Limited number of papers per semester
- Avoid time conflicts

Note: CSPs are not typically for expressing (soft) **preferences** (e.g. minimise difficulty, balance subject areas, etc.)

Example

Choice of Variables & Values

Constraints

1. Pre-requisities*
2. Papers offered on limited semesters-filter
3. Limited number of papers per semester ✓
4. Avoid time conflicts-filter

Variables

A. Semesters?

Domains

- Legal combinations of, for example, 4 papers per semester (but this would result to a **huge set of values**)

- **n – number of choices**
- **r - number of elements per group**
- **No repeats**

n	32
r	4

Formula:

Permutations	863,040	if the paper order matters	**Permutations = (n!)/(n-r)!**
Combinations	35,960	if the paper order does not matter	**Combinations = (n!)/(r!(n-r)!)**

https://www.mathsisfun.com/combinatorics/combinations-permutations.html

Example

Constraints

1. Pre-requisities*
2. Papers offered on limited semesters-filter
3. Limited number of papers per semester ✓
4. Avoid time conflicts*

Choice of Variables & Values

Variables

A. Semesters?

B. Semester Slots?

Subdivide semesters into slots e.g. 4 of them

(Fall 1, 1)(Fall 1, 2)
(Fall 1, 3)(Fall 1, 4)

Domains

- Legal combinations of, for example, 4 papers per semester (but this would result to a huge set of values)
- Papers offered during that semester

8 semesters * 4 slots = 32 variables

Example

Constraints

1. Pre-requisities
2. Papers offered on limited semesters
3. Limited number of papers per semester
4. Avoid time conflicts

Choice of Variables & Values

Variables	Domains
A. Semesters?	• Legal combinations of, for example, 4 papers per semester (but this would result to a huge set of values)
B. Semester Slots?	• Papers offered during that semester
Subdivide semesters into slots (e.g. 4 of them (Fall 1, 1)(Fall 1, 2) (Fall 1, 3)(Fall 1, 4)	
C. Papers?	• Semesters or semester slots (semester slots allow expressing constraint on limited number of papers/ semester)

Example

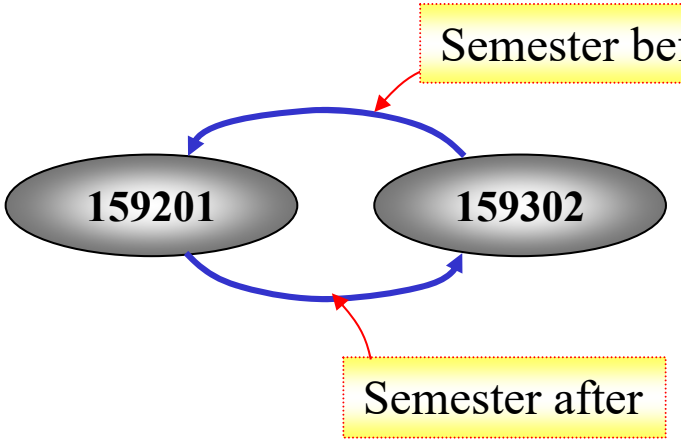
Constraints

1. Pre-requisities *
2. Papers offered on limited semesters
3. Limited number of papers per semester
4. Avoid time conflicts

Constraints

Use papers as variables and semester slots as values.

Prerequisite



- For pairs of papers that must be ordered.

- Complementary constraints

Example

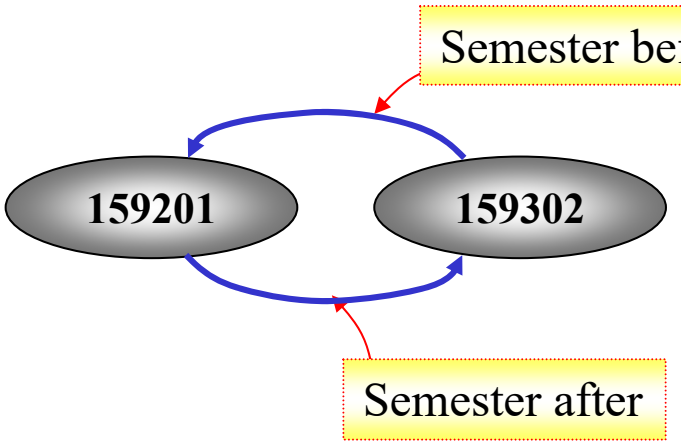
Constraints

1. Pre-requisities *
2. Papers offered on limited semesters-filter
3. Limited number of papers per semester
4. Avoid time conflicts

Constraints

Use papers as variables and semester slots as values.

Prerequisite ⇨



- For pairs of papers that must be ordered.

Papers offered only in some semesters ⇨

- Filter domain

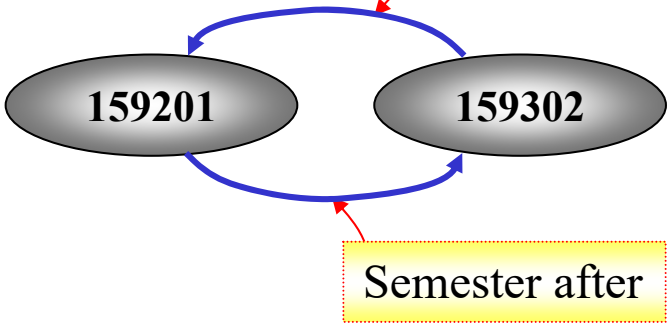
Constraints

Constraints

1. Pre-requisities*
2. Papers offered on limited semesters-filter
3. Limited number of papers per semester*
4. Avoid time conflicts

Use papers as variables and semester slots

Prerequisite ⇨

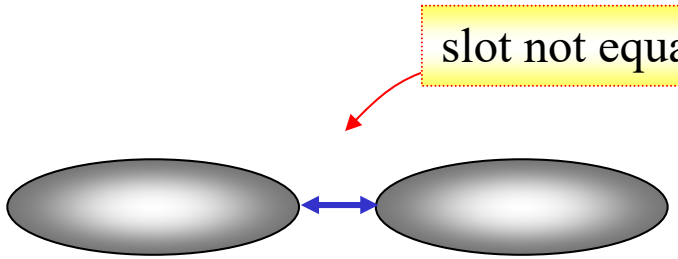


- For pairs of papers that must be ordered.

Papers offered only in some semesters ⇨

- Filter domain

Limit # papers ⇨



- Use semester-slots only once

CSP

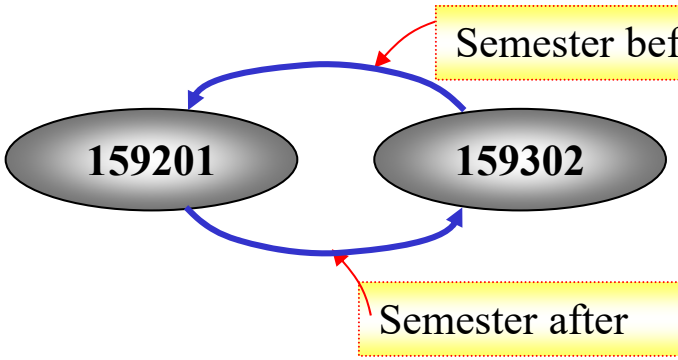
Constraints

Constraints

1. Pre-requisities*
2. Papers offered on limited semesters-filter
3. Limited number of papers per semester*
4. Avoid time conflicts*

Use papers as variables and semester slots

Prerequisite ⇨

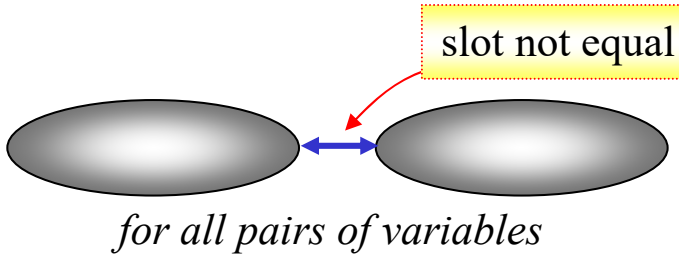


- For pairs of papers that must be ordered.

Papers offered only in some semester s ⇨

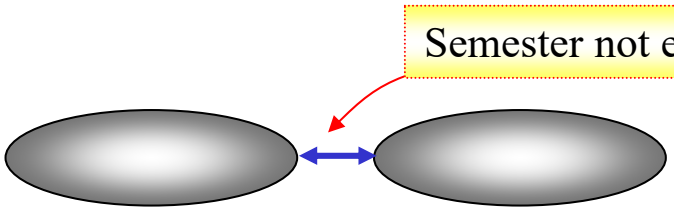
- Filter domain

Limit # papers ⇨



- Use semester-slots only once

Avoid time conflicts ⇨



- For pairs offered at same or overlapping times